

Exercise - Week 9: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Cardinalities, Relations on Sets

Student Name:

Student number: |—|—|—|—|—|—|—|—|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

Problems.

- (1) **(4 marks)** Let $A := \{5k + 2 \mid k \in \mathbb{Z}\}$. Does A have the same cardinality as $\mathbb{Z}$? Justify your answer.

- (2) **(6 marks)** Let ρ be a relation on $\mathbb{Z}$ defined by $x\rho y$ if $a + b$ is even. Determine whether ρ is reflexive, symmetric, or transitive.